Harmonic filters for a six-pulse rectifier

Can a GA scored by geometric-algebra power raise a rectifier's power factor? — milestone M9f, CLAUDE.md §3b item 3

kronspice simulation report · 2026-11-07 · 10 generations × 24, seed 1

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Summary

A three-phase diode bridge draws distorted current from its supply. This example evolves two series L–C filter branches per phase at the point of common coupling (PCC) so that the power factor rises. The fitness is a power-quality measure computed with geometric algebra (GA): the geometric product of the harmonic voltage and current multivectors splits into active, reactive and distortion parts, and names the pairs of harmonics that carry the distortion. The result is checked against IEEE Std 1459 and against the quaternion instantaneous power of the three phases.

- **Power factor 0.9033 → 0.9559** with the default run (10 generations × 24, 222 evaluations); total active power 7389 W → 7673 W.
- **Evolved branches:** L5 = 3.9 mH with C5 = 82 µF (281 Hz) and L7 = 0.15 mH with C7 = 10 µF (4109 Hz) — not at 300 Hz and 420 Hz.
- **Current distortion falls:** D_I 1018 → 784 VA and THD_I 40.4 % → 30.6 % on phase a, while THD_V rises from 9.9 % to 12.8 %.
- **The GA names the culprits:** the largest distortion pair 1–5 falls from 961 VA to 578 VA; fundamental reactive power swings from +515 var to −73 var.
- **Twins agree:** GA and IEEE 1459 per phase to 0.3 % in S; the quaternion mean power equals ΣP (7389 W and 7673 W).
- **A longer run** (30 × 20, 562 evaluations, *doc/M9f.md*) reached PF 0.957 with branches at 284 Hz and 673 Hz.

Background

CLAUDE.md §3b item 3 describes geometric-algebra power theory: for non-sinusoidal waveforms each harmonic's cosine and sine components are basis vectors of a geometric algebra, voltage V and current I become multivectors, and the geometric product M = V·I splits into a scalar (active power) and bivector parts (reactive and distortion power), each tied to a specific pair of harmonics. These measures are meant to feed GA fitness functions for power-quality filters. Milestone M9f builds this in *analysis/power.go* and checks it against IEEE Std 1459-2010; this example is its demonstration. No hypothesis is involved.

The model

- **Supply:** 60 Hz, 120 V (phase), three-phase, with 0.05 Ω and 0.5 mH per line.
- **Load:** a diode bridge into a 2200 µF, 10 Ω DC load.
- **Filter:** at the PCC each phase carries two series L–C branches to a grounded star point, nominally for the 5th and 7th harmonics, each with a fixed branch resistance of 0.05 Ω.
- **Genome:** L5, L7 (0.1–10 mH) and C5, C7 (10–1000 µF), all on the E12 series.
- **Harmonic analysis:** for each phase the last two of 14 simulated cycles are resolved into harmonics up to the 25th (*analysis.HarmonicsOf*).
- **GA power:** the geometric product M = V·I in Cl(2H), with H the number of harmonics (*analysis.GeometricPower*), gives the scalar part P, the same-harmonic bivectors Q and the cross-harmonic bivectors D, grouped by pair of harmonics.
- **Fitness:** 1 − PF, with PF = ΣP/ΣS over the three phases (S the apparent power).

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Results

Default run: 10 generations × 24, 222 evaluations. Per-phase quantities are for phase a. Q_1 is the fundamental reactive power, D_I the current-distortion power of IEEE 1459 ($|V_1 I_H|$), THD total harmonic distortion.

	No filter	Evolved
PF	0.9033	**0.9559**
P (three phases)	7389 W	7673 W
Phase a: S / P / Q_1	2731 VA / 2463 W / +515 var	2701 VA / 2559 W / −73 var
D_I	1018 VA	784 VA
THD_I / THD_V	40.4 % / 9.9 %	30.6 % / 12.8 %
Largest distortion pairs	1–5 961 VA, 1–7 333 VA, 1–11 192 VA	1–5 578 VA, 1–7 461 VA, 1–11 156 VA

Evolved branches: L5 3.9 mH with C5 82 µF (281 Hz) and L7 0.15 mH with C7 10 µF (4109 Hz).

Why the branches are not at 300 Hz and 420 Hz

- Maximizing PF trades harmonic absorption against the leading fundamental reactive power the capacitors add.
- The 5th-harmonic branch is tuned a little low, which removes most of the 1–5 distortion and swings Q_1 from +515 var to −73 var.
- The second branch in this run is mostly a capacitor.

Longer run (doc/M9f.md)

Run	PF	THD_I	Largest distortion pairs (phase a)
No filter	0.903	40 %	1–5 961 VA, 1–7 333 VA, 1–11 192 VA
Evolved, 30 × 20, 562 evaluations	0.957	28 %	1–5 611 VA, 1–7 402 VA, 1–13 151 VA

Its branches sit at 284 Hz and 673 Hz, Q_1 goes from +515 var to −226 var per phase, and the quaternion mean power equals ΣP (7389 W and 7736 W). The one-line summary in examples/README.md ("PF 0.903 to 0.957") matches this longer run.

Validation

In this example

- The GA and IEEE 1459 columns agree per phase, S to 0.3 % (the commutation notches above the 25th harmonic are outside the GA's harmonic set).
- The quaternion instantaneous power of the three phases (scalar part of v·i*) averages to ΣP: 7389 W and 7673 W.
- *main_test.go*: GA against IEEE 1459, largest pair 1–5, quaternion mean = ΣP, and a short GA run that must improve PF by at least 0.01. The program exits non-zero if the evolved filter does not improve the power factor.

Checks of the GA power code (doc/M9f.md)

Check	Result
GA vs IEEE 1459 on random band-limited waveforms (9 harmonics, DC offset): P, S, S_1, D_I, D_V, S_H, P_1, Q_1; $S^2 = P^2 + Q^2 + D^2$; $S^2 = S_1^2 + D_I^2 + D_V^2 + S_H^2$; Σ pair² = D^2	50 cases, 10^{-9}
R–L from the engine's transient: P = I²R, Q_1 = I²ωL, no distortion	10^{-4}
Single-phase bridge rectifier from the engine: GA and IEEE 1459 agree (S to 1 %: harmonics above the 32nd); largest distortion pair $V_1 \times I_3$	PF 0.56, THD_I 140 %
Quaternion power s = v·i*: scalar Σv·i, vector −v×i; p invariant and vector co-rotating under Clarke and Park	10^{-13}, 10^{-10}
Clarke and Park rotors = matrices	500 random angles and vectors, 10^{-14}

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Exported circuits

export_test.go writes the unfiltered rectifier and one with 1 mH, 270 µF / 150 µF branches (300 Hz, 420 Hz) as netlists, KiCad schematics and results sheets in *export/*. Kron's torn solution is shown there for linear circuits only; this circuit contains diodes, so the sheet reports the MNA operating point alone (the torn Newton solve is checked in kron's tests).

Limits of the model

- Harmonics are resolved only up to the 25th, so commutation notches above it are left out of the GA power (the 0.3 % gap in S).
- IEEE 1459 three-phase effective quantities (S_e, positive-sequence P_1^+, unbalance power) and GA power for three-phase sets are not done (doc/M9f.md, "Not done here").
- The supply, rectifier and branches are ideal apart from the stated resistances; the fitness is PF only.

Reproducing the results

```
go run ./examples/power_filter    # about 2 min; -gens 10 -pop 24 -seed 1 are the defaults
go test ./examples/power_filter   # GA and IEEE 1459 agree; a short GA improves PF (-short skips the GA)
KRONSPICE_EXPORT=1 go test ./examples/power_filter -run TestExport
```

Sources: IEEE Std 1459-2010; M. Castro-Núñez, R. Castro-Puche, "The IEEE Standard 1459, the CPC power theory, and geometric algebra in circuits with nonsinusoidal sources and linear loads," *IEEE Trans. Circuits Syst. I*, 2012; doc/M9f.md.

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